

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim : Simulate cross correlation and autocorrelation on discrete time signals	
Lab Experiment : 3	Date: 31/07/2026	Enrolment No: 92400133125

Aim : Simulate cross correlation and autocorrelation on discrete time signals.

Code :

```
import numpy as np

import matplotlib.pyplot as plt

def cross_correlation(signal1, signal2):

    # Compute the cross-correlation

    cross_corr = np.correlate(signal1, signal2, mode='full')

    return cross_corr

def autocorrelation(signal):

    # Compute the autocorrelation

    auto_corr = np.correlate(signal, signal, mode='full')

    return auto_corr

# Define the discrete-time signals

signal1 = np.array([1, 2, 3, 4, 5])

signal2 = np.array([2, 4, 6, 8, 10])

# Compute the cross-correlation

cross_corr = cross_correlation(signal1, signal2)

# Compute the autocorrelation

auto_corr = autocorrelation(signal1)

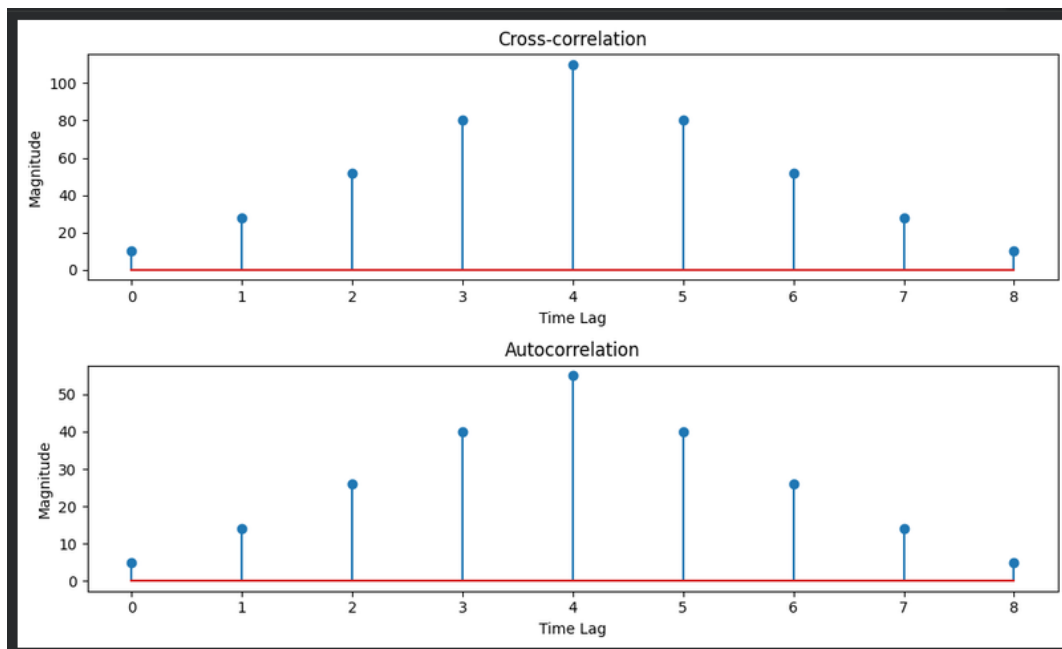
# Plot the cross-correlation and autocorrelation signals

plt.figure(figsize=(10, 6))
```

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```
plt.subplot(2, 1, 1)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.subplot(2, 1, 2)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.tight_layout()
plt.show()
```

Output :



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Conclusion :

The implementation is able to compute the cross-correlation and autocorrelation of the provided signals. The cross-correlation provides information about the degree of similarity between two distinct signals at various time shifts, whereas the autocorrelation indicates how well the signal is similar to itself. As signal2 is scaled from signal1, there is a high peak for cross-correlation, and maximum value occurs at zero lag for autocorrelation.

Calculation :

Experiment 3 Calculation

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$x(n) = [1, 2, 3, 4, 5]$ $n = 0, 1, 2, 3, 4$
 $y(n) = [2, 4, 6, 8, 10]$ $n = 0, 1, 2, 3, 4$

→ Cross - Correlation

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n) y(n-l)$$

l	Shifted signal $y(n-l)$	overlapping sum calculation	$r_{xy}(l)$
-4	[0, 0, 0, 0, 2, 4, 6, 8, 10]	1×10	10
-3	[0, 0, 0, 2, 4, 6, 8, 10, 0]	$2 \times 8 + 2 \times 10$	28
-2	[0, 0, 2, 4, 6, 8, 10, 0, 0]	$3 \times 6 + 2 \times 8 + 3 \times 10$	52
-1	[0, 2, 4, 6, 8, 10, 0, 0, 0]	$4 \times 4 + 2 \times 6 + 3 \times 8 + 4 \times 10$	80
0	[2, 4, 6, 8, 10, 0, 0, 0, 0]	$2 \times 2 + 2 \times 4 + 3 \times 6 + 4 \times 8 + 5 \times 10$	110
1	[0, 2, 4, 6, 8, 10, 0, 0, 0]	$2 \times 2 + 3 \times 4 + 4 \times 6 + 5 \times 8$	80
2	[0, 0, 2, 4, 6, 8, 10, 0, 0]	$3 \times 2 + 4 \times 4 + 5 \times 6$	52
3	[0, 0, 0, 2, 4, 6, 8, 10, 0]	$4 \times 2 + 5 \times 4$	28
4	[0, 0, 0, 0, 2, 4, 6, 8, 10]	5×2	10

Cross - Correlation = [10, 28, 52, 80, 110, 80, 52, 28, 10]

Auto - Correlation

$$R_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n) x(n-l)$$

l	Calculation	$R_{xx}(l)$
-4	1×5	5
-3	$1 \times 4 + 2 \times 5$	14
-2	$1 \times 3 + 2 \times 4 + 3 \times 5$	26
-1	$1 \times 2 + 2 \times 3 + 3 \times 4 + 4 \times 5$	40
0	$1^2 + 2^2 + 3^2 + 4^2 + 5^2$	55
1	$2 \times 1 + 3 \times 2 + 4 \times 3 + 5 \times 4$	40
2	$3 \times 1 + 4 \times 2 + 5 \times 3$	26
3	$4 \times 1 + 5 \times 2$	14
4	5×1	5

Auto-Correlation: [5, 14, 26, 40, 55, 40, 26, 14, 5]